



International Journal of Research Publication and Reviews

Journal homepage: www.ijrpr.com ISSN 2582-7421

PROTECTION OF TRADITIONAL KNOWLEDGE IN INDIA: AN ANALYTICAL STUDY

Ruchika¹, Dr. Maya Devi²

L.L.M (Master of Laws), University Institute of Legal Studies (Department of Law), Chandigarh University, Mohali, Punjab, India.

Assistant Professor, University Institute of Legal Studies (Department of Law), Chandigarh University, Mohali, Punjab, India.

ABSTRACT :

Traditional knowledge is the old knowledge and practice which come from old time of our ancestors. In India, this knowledge is very important because it is part of culture, custom, and day-to-day life of local people. It includes use of herbs, farming system, healing method, and many other natural ways which people use from many generations. But in present time, this knowledge is not safe because many foreign company and person try to take it and make profit without giving any benefit to the community who own it. Such misuse is called biopiracy and it became big problem for India.

The government of India make many steps to protect traditional knowledge like making Traditional Knowledge Digital Library (TKDL), Biological Diversity Act, 2002 and Geographical Indication Act, 1999. These law and system try to stop the wrong use of traditional knowledge and give some right to local people. But still, there is no one strong law made only for this purpose. Many poor and tribal people not know their right and their knowledge get used without permission. This study tries to understand how India is protecting traditional knowledge now, what are the gap in the system, and what more step should be taken to make it strong. It also looks how international rule like Convention on Biological Diversity and Nagoya Protocol can help India in this matter.

This research tries to find new way and suggestion for better protection of traditional knowledge and fair benefit for the real holder of this wisdom.

Keywords: Traditional Knowledge, Biopiracy, Protection, TKDL, Biodiversity Act, Geographical Indication, Indigenous Community, Intellectual Property.

INTRODUCTION

Traditional knowledge is old wisdom come from ancestors and passed from one generation to another. It is knowledge of farming, medicine, culture, art, and environment which local

people keep safe in their mind and practice. In India, this knowledge is part of daily life of village people and tribal communities. But now many big company and foreign people use this knowledge for their business without giving any respect or money to the real owner.

Because of this, many traditional people lose their rights and benefit. India has many laws and steps for protect traditional knowledge like TKDL and patent rules, but still there are many problems and gap in system. The protection of traditional knowledge is not only for save old culture, but also to make sure people who keep this knowledge get fair benefit and respect.¹

Research Questions

The research questions for the study are as follows: -

1. What are the main law and policy in India for protection of traditional knowledge?
2. How effective is the Traditional Knowledge Digital Library (TKDL) in stop misuse of Indian traditional knowledge?
3. What are the main challenges and gap in present legal system for traditional knowledge protection?
4. How can India make better and strong system for protect traditional knowledge and local community rights?

Problem Statement

In India, traditional knowledge is important part of heritage, but there is no full and clear law for protect it. Many company and foreign people take this knowledge and make patent without give credit to real people. The present system like TKDL and intellectual property laws are not enough to stop misuse and biopiracy. Local people do not have awareness or legal power to protect their own knowledge. There is big need to study how the system work and what change can make it stronger for protect India's traditional knowledge and help community to get justice and recognition.

Objectives of the Research

1. To study the meaning and importance of traditional knowledge in India and its role in daily life of local people.
2. To find out the legal system and policy made by government for protection of traditional knowledge.
3. To examine how Traditional Knowledge Digital Library (TKDL) help in stop misuse and biopiracy of Indian knowledge.
4. To identify the problem and weakness in present law and suggest better way for strong protection.
5. To understand how protection of traditional knowledge can give benefit and respect to local and tribal communities.

History and Evolution of Traditional Knowledge Protection in India

Traditional knowledge in India is very old and come from long time before modern science and technology start. It is knowledge of how to live with nature, how to cure people with

herbs, how to grow crop, and how to make art and craft. This knowledge come from ancestors and go to next generation through teaching and practice. In old days, people never write this knowledge in books but keep it in their mind and tell it to their children. This makes it part of cultural and social life of every community. The old Indian text like Vedas, Ayurveda, Yoga, and Unani medicine are big example of traditional knowledge which show how deep Indian people understand nature and human health.²

In ancient time, traditional knowledge was protected by custom and community rule. The people believe this knowledge belong to everyone in the community and no one can own it alone. The village elders and traditional healers keep it safe and pass it only to trusted people. There were respect and moral duty to not misuse the knowledge for wrong work. But when British rule come in India, they bring new law and system which not understand Indian traditional system. The western patent and copyright law look only for written record and inventors, not for community knowledge. Because of this, traditional knowledge loses its protection and many foreign people start taking Indian knowledge and use it for making business and patent.

After independence, India starts to understand that its traditional knowledge is being stolen and used by other country without permission. Many examples like neem patent case, turmeric case, and basmati rice case show how foreign company take Indian traditional knowledge and try to make profit. These cases open eyes of Indian government and public that there is need for special system to protect old knowledge. Slowly India begins to take step for make proper law and digital record. The Council of Scientific and Industrial Research (CSIR) make big effort to stop biopiracy and start project to collect traditional medicine knowledge.³

In 2001, India creates Traditional Knowledge Digital Library (TKDL) to protect old knowledge from being patented by foreign people. TKDL collect old Indian books and text on Ayurveda, Unani, Siddha, and Yoga, and translate them in different language so patent offices in other country can check before give any patent. This system helps to stop many wrong patents on Indian knowledge. It became one of the most important steps in protection of traditional knowledge at world level. The TKDL show that traditional knowledge can also be protected in modern way by using technology and record keeping.⁴

After that, India also makes law and policy for benefit sharing and biodiversity. The Biological Diversity Act, 2002 help to protect knowledge related to plants, herbs, and animal and make rule that if anyone use this knowledge, the local community should get share in benefit. India also supports many international agreements like Convention on Biological Diversity (CBD) and Nagoya Protocol which talk about fair use and sharing of benefit from traditional knowledge.⁵

Still, protection of traditional knowledge in India is not complete. Many problems exist like lack of awareness among local people, poor record keeping, and weak legal framework.

Many communities not know their rights and many company still try to use this knowledge without proper permission. There is also confusion between intellectual property law and traditional knowledge because traditional knowledge is collective and old, not new invention.

But with time, government, researcher, and people are trying to make strong system. Many universities and research centre start work on traditional knowledge and its documentation. The new digital tools, awareness program, and law-making help to make system better. The world is also now giving more value to traditional knowledge for sustainable development, health, and environment.

So, the journey of traditional knowledge protection in India go from oral tradition to digital record. From old village healer to modern legal framework, India has come long way. But still there is need for sui generis law, more awareness, and proper benefit sharing system to give full respect and justice to the keeper of this knowledge. The history and evolution show that traditional knowledge is not only part of past but also important for future of India.

Meaning and Importance of Traditional Knowledge

Traditional knowledge means the old wisdom, skill, and practice which come from ancestors and pass from one generation to another. It is not written in book but live in mind and work of people. It includes knowledge about plant, animal, medicine, farming, art, food, and natural living. It is part of daily life of people in village and tribal area who use it for health, farming, and making of useful things. This knowledge is not belonged to one person but to whole community who protect and share it with respect.⁶

Traditional knowledge is very important for Indian people because it show their culture, history, and way of life. It helps in understanding how people live close to nature and take care of environment. It is also base of many Indian systems like Ayurveda, Yoga, and local art which make India rich in

heritage. It gives identity to community and helps them earn their living by using their skill. This knowledge also supports sustainable living because it teaches how to use natural resources without destroy it.

In modern time, traditional knowledge becomes more valuable because it helps in making medicine, organic farming, and natural product. Many companies use this knowledge for business, but it is important that people who keep this knowledge get fair respect and benefit. So, traditional knowledge is not only part of old culture, but also big strength for India's growth, environment care, and cultural pride.

Types and Forms of Traditional Knowledge in India

Traditional knowledge in India is of many types and forms which come from long time of living and experience of people. It is knowledge made by village and tribal people who learn from nature and from their old generation. Every part of India has its own kind of knowledge which tell about life, culture, and work of that place. This knowledge is used for health, food, farming, building, and many daily needs of people. It is simple but full of deep understanding about how to live in balance with nature.

One big type of traditional knowledge is **medicine and healing**. It includes system like Ayurveda, Unani, Siddha, and local herbal treatment. People use plant, root, and natural thing for curing many diseases. This knowledge is old and still useful for many people who live in village and forest.⁷

Another important type is **farming and agriculture knowledge**. Farmers know how to grow crop, save seed, and take care of soil and water. They follow method which protect nature and give healthy food. This traditional farming help in making sustainable and chemical-free life.⁸

Traditional **art and craft** are also big part of Indian knowledge. People make handcraft, weaving, painting, and pottery with old skill which show beauty and tradition. Each region has its own style and design which tell story of their culture and belief.⁹

Traditional knowledge is also strong in **environment and natural resource management**. People know when to sow seed, how to save forest, and how to use water wisely. They respect nature like family and do not harm it.¹⁰

So, the types and forms of traditional knowledge in India are many and cover every part of life. It shows the wisdom, creativity, and strong connection between human and nature which make India rich in culture and knowledge.

Role of Traditional Knowledge Digital Library (TKDL)

Traditional Knowledge Digital Library (TKDL) is made by India to protect old knowledge from being stolen or misused by foreign people and company. It is like a big record where Indian traditional knowledge is written and kept safe in digital form. The main aim of TKDL is to stop wrong patent on Indian traditional medicine and to show that this knowledge already exists in India from long time. It collects information from old book and text of Ayurveda, Unani, Siddha, and Yoga and translate them in different language so that foreign patent office can understand it.¹¹

TKDL help to stop biopiracy and protect Indian knowledge from being claimed by other country. Many time, foreign company try to take patent on product made from neem, turmeric, or other Indian herbs. Because of TKDL, now patent office can easily check and cancel such wrong application. It gives proof that this knowledge is not new invention but part of old Indian system.

The TKDL also help in creating awareness among people and government about importance of traditional knowledge. It shows that even old knowledge can be protected in modern way by using technology. It is example how India can mix tradition with science for better result. TKDL also support international cooperation by sharing information with other country patent office under agreement.

This system protects not only the knowledge but also the respect of Indian people who keep it safe for generation. It stops misuse and helps community to get fair right. Still there is need to make TKDL bigger by adding more local knowledge and train people to use it properly. In this way, TKDL play very big role in saving India's heritage and giving strong protection to traditional knowledge at global level.

Literature Review

The study of traditional knowledge and its protection in India have been discussed by many scholars, writers, and researchers who show how important it is for culture, identity, and development. Different studies talk about how traditional knowledge is part of daily life of people and how it faces problem from modern law and misuse by company. The review of literature help to understand what work has already been done and what still need to be studied for better protection system.

Many scholars like **Anil K. Gupta (1999)** write that traditional knowledge is the base of local innovation and must be protected by giving respect and reward to the community. He explains that people who keep this knowledge are poor in money but rich in wisdom. **R. D. Gupta (2003)** and others show that Indian farming and healing system have deep scientific base which can help in sustainable living if used in right way. They also tell that without strong law; many foreign companies take Indian knowledge and make patent for profit.¹²

Other researcher like **Nirmal Sengupta (2007)** say that the main problem is that western intellectual property law does not fit with traditional system. Traditional knowledge is **R. S. Chauhan (2010)** write that India has made effort through the Biological Diversity Act, 2002 and creation of Traditional Knowledge Digital Library (TKDL) to stop biopiracy. He also mentions that TKDL has helped India to cancel many wrong patents in foreign country. Studies by **Pushpendra Kumar and Meera Patel (2015)** also support that TKDL is one of the most successful examples in the world for protection of traditional medicine.¹⁴

Some studies like **Shamnad Basheer (2011)** focus on the need for sui generis law in India. They argue that present intellectual property laws are not enough to protect collective and oral knowledge. They suggest that India must make a special law that understand community rights and traditional system.¹⁵

Ritu Priya and S. K. Sharma (2016) study how local people and traditional healers are not aware about their rights and how government should make awareness programs for them. They say that protecting traditional knowledge is not only legal matter but also social and ethical duty.¹⁶

Another study by **UNESCO (2018)** show that traditional knowledge is important for biodiversity, climate change, and sustainable development. It tells that local people are the best protector of nature because of their long experience and understanding.¹⁷

Recent research by **WIPO (2020)** discuss how global efforts are being made to include traditional knowledge in international intellectual property system. It says that India is one of the leading countries in this field but still need more work for benefit sharing and legal recognition.¹⁸

From all these studies, it is clear that traditional knowledge has great value and need strong protection system. Many researchers agree that present law is not enough and community- based approach is required. They also show that documentation like TKDL is useful, but it should also ensure that people who hold the knowledge get fair benefit.

Thus, literature review show that while India has made progress in protecting traditional knowledge, many gaps remain. There is still need for more study, awareness, and strong legal system to make sure that traditional knowledge is protected, respected, and used for the benefit of real owner and future generation.

Legal Framework for Traditional Knowledge in India

India has made many laws, rule, and policy to protect traditional knowledge from misuse and biopiracy. The government understand that old knowledge of people is valuable and must be kept safe under legal system. Traditional knowledge cannot be protected fully under normal intellectual property law like patent or copyright because it belongs to whole community and come from long time. So, India tries to make special way to give protection through different act and policy.

One main law is **Biological Diversity Act, 2002**. This law is made to protect the knowledge related to plant, animal, and other living thing. It says that if any person or company want to use this knowledge for research or business, they must take permission and give benefit to local people. This act helps to stop foreign people from taking India's bio-resource and traditional knowledge without sharing profit. It also makes rule for fair use and conservation of biodiversity.¹⁹

Another big step is **Traditional Knowledge Digital Library (TKDL)** which is not a law but a strong system made by government to protect knowledge of Ayurveda, Unani, Siddha, and Yoga. It collects old information and records it in computer system so that no foreign patent office can give wrong patent on Indian knowledge. TKDL help in many cases like neem and turmeric patent which show that this knowledge is already part of India.²⁰

India also follows some international agreement like **Convention on Biological Diversity (CBD)** and **Nagoya Protocol**, which say that country must protect traditional knowledge and

Even with all these efforts, many problems still remain. There is no one full law only for traditional knowledge, and many people in village do not know their rights. The system needs to be simpler and stronger to make sure real owner of knowledge get respect and benefit. The legal framework of India is trying to grow step by step to protect old wisdom and make it safe for future generation.

International Efforts and Agreements

Protection of traditional knowledge is not only work of one country but also matter for whole world. Many international effort and agreement are made to stop misuse of local knowledge and to give fair benefit to people who keep it. These efforts help country like India to make strong law and work together with other nation for protection of old wisdom.

One of the main agreements is **Convention on Biological Diversity (CBD)**, which was made in 1992. This agreement talks about how to protect biodiversity and the knowledge related to it. It says that traditional knowledge of local and tribal people is very important for saving nature and must be used only with their permission. It also tells that when any company or person use this knowledge for making product, they must share benefit and respect the rights of the people. India is member of this convention and follow its rule in national law.²³

Another big agreement is the **Nagoya Protocol**, which was made in 2010 under CBD. It gives clear rule for **Access and Benefit Sharing (ABS)**. It means that no one can take or use genetic resources or traditional knowledge without consent of the community or country. If they use it, they must give fair money or benefit back to the people who own it. The Nagoya Protocol help to make international trade and research fairer and more transparent. India also joins this protocol and makes rule under Biological Diversity Act to follow it.²⁴

The **World Intellectual Property Organization (WIPO)** also work for protection of traditional knowledge at global level. WIPO make one special group called **Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore (IGC)**. This group talk with different country to make international rule for protect traditional knowledge, traditional culture, and genetic resources. WIPO help country to share idea, make database, and plan new system for fair protection.²⁵

Many times, India takes active part in WIPO meeting and tells that traditional knowledge cannot be treated same as normal patent or copyright. India support idea of **sui generis system**, a special law made only for traditional knowledge.

So, these international effort like CBD, Nagoya Protocol, and WIPO work show how the world is trying to make strong and fair system to protect traditional knowledge. They give base for country like India to make their own rule and ensure that people who keep old wisdom get respect, right, and benefit for their contribution to the world.

Case Studies of Biopiracy in India

India has faced many cases of biopiracy where foreign company take traditional knowledge and make patent without give any respect or benefit to real owner. These case show how Indian old wisdom and natural resource are misused for business. Some famous examples like neem, turmeric, and basmati rice case help India to understand the need of protection for traditional knowledge.

The first big case is the **Neem case**. Neem tree is used in India from ancient time for medicine, farming, and pest control. People use neem oil and leaf for many purposes like skin disease, insect control, and health care. But a foreign company get patent in the United States for using neem as pesticide. Indian people already know this use from many years, so this patent was wrong. Indian organization and research group fight against it and show proof that neem use is old Indian knowledge. Finally, the patent was cancelled, and it became big victory for India.²⁶

The second famous case is the **Turmeric case**. Turmeric is common spice in every Indian home and used for healing wound, skin care, and many diseases. Two researchers from a foreign university get patent in America for use of turmeric in healing property. Indian scientist from Council of Scientific and Industrial Research (CSIR) prove that this is already known in India from long time. They give old text and record as proof, and the patent was cancelled. This case show how Indian traditional knowledge can be protected with proper documentation²⁷.

Another important case is the **Basmati rice case**. A foreign company from United States take patent for a kind of rice similar to Indian basmati and try to sell it in other country. Basmati is special rice grown in India and Pakistan from many years. Indian government and farmer protest because it was part of their traditional farming knowledge. After long fight, the claim was removed, and India save the name and pride of its traditional product.²⁸

These case show that traditional knowledge is always at risk of being taken by people who only want profit. They also show how India must protect its knowledge through strong law and system like TKDL. These examples become lesson for the world that old community knowledge must be respected and not used for wrong patent. The neem, turmeric, and basmati case make big change in Indian legal and public awareness about importance of protecting traditional knowledge for future generation.

Challenges in Protection of Traditional Knowledge

In India, there are many challenges in protection of traditional knowledge even after having law and system. The main problem is that traditional knowledge is old, collective, and passed by word of mouth, not written in book. Because of this, it is very hard to prove who is the real owner. Many people and company take this knowledge and use it for making product and profit without permission. The community who keeps this knowledge get no respect or benefit for their wisdom.

One big challenge is **lack of awareness** among local and tribal people. Many of them do not know that their old knowledge has legal value and can be protected. They also not understand how to claim their right under law. This makes it easy for others to misuse their knowledge.

Another problem is **weak legal system**. The law in India like Biological Diversity Act and Patent Act give some help, but there is no one full law made only for traditional knowledge. The normal intellectual property law not fit well because they are made for new invention and single owner, not for community and old practice. So, many traditional knowledge case cannot get full justice.

There is also issue of **poor documentation and record keeping**. Much of the knowledge is still not written or collected. Only few parts like Ayurveda and Unani are recorded in TKDL, but many local and tribal knowledge remain unprotected. Without record, it becomes easy for company to take it and make patent in foreign country.

Misuse by company and foreign organization is another big challenge. Many time they take plant, herbs, and local method and claim it as their own invention. They make profit from it but never share with people who actually gave the knowledge. This is called biopiracy, and it harm both culture and economy of India.²⁹

Other problem includes lack of government coordination, shortage of trained officer, and slow action in court case. Also, young generation is not showing much interest in keeping old knowledge alive because they think it is not useful in modern life.

So, the challenges in protection of traditional knowledge are many and serious. There is need of strong law, more awareness, better record system, and active role of government to make sure that traditional knowledge is safe and respected. Only then the wisdom of old people and culture of India can be kept alive for future generation.

Role of Community and Local People in Safeguarding Knowledge

Community and local people play very big role in keeping and protecting traditional knowledge in India. They are the real owner and guardian of this knowledge, which come from their ancestors and daily life. In every village and tribal area, people learn from nature and from old generation about farming, healing, food, and craft. They do not write this knowledge in book but remember it by practice and sharing. Their life, culture, and belief are strongly connected with this knowledge.³⁰

The **village elders and traditional healers** are main source of old wisdom. They teach younger people about herbs, plant, and treatment for different sickness. They also tell them the right time for farming, how to save seed, and how to take care of soil and water. By doing this, they make sure the knowledge does not die with them but pass to next generation.³¹

Tribal communities are also very important for protecting traditional knowledge about forest and nature. They know which plant is used for food, which for medicine, and which for building. They use natural resources carefully without destroy it. Their method of living show how to live in balance with nature. They also follow their own rule and tradition to keep the knowledge safe from outsider.

Local people also help in **identifying and recording** their knowledge when government or researcher come to make documentation. They share their experience and talk about the old method of using natural resources. Without their help, no project like TKDL or Biodiversity Board can collect true information.

The community also act as **protector from misuse**. They stop outsider from taking their plant, herbs, or design without permission. Many local group and NGO help them to understand their right and to fight against biopiracy.

But still, these people need more support and awareness from government and society. They should get fair benefit when their knowledge is used for research or business. The role of community is not only to keep old practice but also to protect it with pride and teach it to coming generation.

So, the village and tribal people are the heart of traditional knowledge in India. They are living library of wisdom, and their active role is most important to safeguard and keep this knowledge alive for the future of the country and the world.

Need for Sui Generis Law for Traditional Knowledge in India

There is big need for special and new law only for protection of traditional knowledge in India. The normal intellectual property law like patent, copyright, and trademark cannot give full protection because traditional knowledge is very different from modern invention. It belongs to whole community, not to one person, and it come from long time of practice and experience.

Because of this, India needs **sui generis law**, which mean a separate and unique law made specially for traditional knowledge.³²

Traditional knowledge is passed from one generation to another through oral way, not written in document. It is part of daily life, culture, and belief of people. The present law system cannot understand this kind of knowledge because it only protects new and individual creation. So, many time foreign company take Indian knowledge, make patent, and earn profit. The local people get nothing in return. A sui generis law can solve this by giving legal right to community and by stop misuse or biopiracy.

Such law can also help in **benefit sharing**, which mean if someone use traditional knowledge for making product or research, they must give fair money or benefit to the people who own it. It can also make system for **documentation and registration** of traditional knowledge in safe way, without losing control of the community over it.³³

A special law will also make clear **definition of ownership** and protect both written and unwritten knowledge. It can give power to local and tribal people to claim their right in court when their knowledge is stolen. It can also support India in world level discussion under WIPO and other international body for protection of traditional knowledge.

Many experts and researcher already say that only a sui generis system can truly match the nature of traditional knowledge. It can mix traditional value with modern law and make fair system for everyone.

So, India must make one strong sui generis law for protection of traditional knowledge. It will help to give legal power, fair benefit, and respect to local people who keep this old wisdom alive. This law will protect India's cultural heritage and make sure that traditional knowledge is not misused but used in right way for future generation.

Benefit Sharing and Economic Value of Traditional Knowledge

Traditional knowledge has great economic value and can give fair benefit to local and tribal people if used in right way. The knowledge of medicine, farming, art, and craft kept by village people is not only part of their life but also useful for big company and researcher who make

new product from it. When this knowledge is used for making medicine, cosmetic, or organic product, the people who keep it must get fair share of money and respect. This is called **benefit sharing**.

Benefit sharing mean giving proper reward to community when their knowledge help in making business or research. It can be in form of money, job, training, or development of their village. For example, if a company use one herbal formula known by local healer, they must pay some part of profit to that healer or his community. This help to make system fair and stop misuse of knowledge.

The **Biological Diversity Act, 2002** in India already talk about benefit sharing. It says that no one can use plant, animal, or traditional knowledge for profit without permission and sharing benefit with people who own it. The National Biodiversity Authority and State Board make sure that local people get their right when their knowledge is used.³⁴

Traditional knowledge also has high **economic value** in global market. Many people in other country want natural and traditional product like Ayurvedic medicine, handmade craft, or organic food. This creates new business chance for Indian community. If government help them with training, marketing, and fair-trade system, they can earn good income and live better life.³⁵

Benefit sharing also give **social respect** to local people. It shows that their knowledge is not poor or old but useful for whole world. It also motivates young generation to learn and keep their traditional knowledge alive.

But in many case, local people not get proper benefit because of weak law, lack of awareness, and middleman. There is need of strong rule, fair agreement, and honest system to make sure benefit go to real owner.

So, benefit sharing and economic value of traditional knowledge are very important for protection and growth of community. It makes justice to people

who keep old wisdom and also support development, equality, and respect for their cultural heritage.

Future of Traditional Knowledge Protection in India

The future of traditional knowledge protection in India depends on how the country make strong law, better policy, and awareness among people. India has rich and old knowledge system, but it needs more effort to keep it safe from misuse and to give fair benefit to community. The next step should focus on making system that understand both modern law and traditional value.

First, India needs to make one **sui generis law** only for traditional knowledge. The normal patent and copyright law cannot protect community-based knowledge, so a new law is required to give clear right and rule for ownership, use, and benefit sharing. This special law can help to stop foreign company from taking Indian knowledge and making profit without permission.³⁶

Second, there must be more **awareness programs** for village and tribal people. They should know that their old knowledge has value and legal right. Government, NGO, and school can work together to teach them how to protect and register their knowledge.

Third, India should make **better documentation system**. Many local knowledge are still not recorded and can be lost or misused. The Traditional Knowledge Digital Library (TKDL) should be expanded to include more regional and tribal knowledge. Modern technology like database and mobile app can be used for easy recording and sharing of information.

Fourth, there should be strong **benefit sharing system**. When company or research use traditional knowledge, the real owner should get fair share of profit. This will encourage community to keep their knowledge alive and use it in safe way.³⁷

Fifth, India should take active part in **international discussion** under WIPO and other global organization. It should continue to fight for fair rule at world level for protection of traditional knowledge and against biopiracy. Cooperation with other country will also help to make stronger system.³⁸

Finally, the future of traditional knowledge protection depends on **young generation**. They must learn from elders and respect their wisdom. Education system can include traditional knowledge study to keep it alive.

So, the future of traditional knowledge protection in India is bright if right steps are taken. With strong law, awareness, documentation, and fair benefit sharing, India can become world leader in protecting and promoting its old wisdom for future generation and sustainable development.

Conclusion

In conclusion, traditional knowledge is very important part of India's culture and life of local people. It is not only old idea but also living wisdom that give help in health, farming, art, and environment. The study shows that this knowledge face big danger from misuse and biopiracy because many company and people use it without giving respect or benefit to real owner. India has made many steps like Traditional Knowledge Digital Library (TKDL) and Biodiversity Act to protect it, but still there are many problems like weak law, lack of awareness, and poor record system. There is strong need of one special **sui generis law** only for traditional knowledge so that it can give clear rule about ownership, protection, and benefit sharing. The local and tribal people should be given more chance to take part in policy and decision about their knowledge. Government must give them training, help, and fair reward when their knowledge is used by company or research. Traditional knowledge is also important for **sustainable development** because it teach people how to live in balance with nature. It can help in solving modern problem like climate change, pollution, and health issue by using natural and simple way. The protection of traditional knowledge also helps to keep cultural diversity and national pride of India. In future, India must make system stronger with help of new technology, good law, and education. Young generation should learn and keep this knowledge alive. If all people, government, and world organization work together, traditional knowledge can become power for development and respect for local community. So, protecting traditional knowledge is not only legal duty but also moral and cultural responsibility for all.

Suggestions and Recommendations

1. **Make new special law:** India should make one special *sui generis* law only for traditional knowledge. The present law like patent or copyright not fully protect it. The new law must clearly say who own the knowledge, how it can be used, and how benefit should be shared.
2. **Increase awareness among people:** Many village and tribal people not know their rights. Government should start awareness camps, school programs, and local meeting to tell them about importance of protecting their knowledge. This will help them stop others from

misusing it.

3. **Strengthen TKDL system:** The Traditional Knowledge Digital Library should be expanded to include more local and regional knowledge. It must be made in simple language so that even local people can understand. Regular update and modern technology should be used for safety and easy access.
4. **Fair benefit sharing:** When company or research use traditional knowledge, the people who give that knowledge must get fair money, job, or development help. The National Biodiversity Authority should check that benefit reach to real owner and not to middleman.
5. **Support from government:** The government should help local people with training, research, and small business support so they can use their knowledge for earning in fair way. More funds and schemes should be made for community-based knowledge projects.
6. **Encourage youth and education:** Traditional knowledge should be taught in schools and colleges so that young generation can learn and respect their heritage. It will also stop old knowledge from dying.
7. **International cooperation:** India should work with other countries and world organizations like WIPO, WTO, and CBD to make strong global rule against biopiracy and unfair patenting.
8. **Proper documentation:** There should be a national system for recording and saving local knowledge before it is lost. Village-level committees can help in collecting and registering the information safely.
9. **Community participation:** Local and tribal people should be included in decision- making for laws and projects related to their knowledge. Their voice and consent must be respected before using their wisdom.
10. **Strict punishment for misuse:** Strong legal action must be taken against any person or company who use traditional knowledge without permission. This will create fear and stop future misuse.

References and Bibliography Books:

1. Gupta, Anil K. *Intellectual Property Rights and Traditional Knowledge: The Indian Experience*. New Delhi: Sage Publications, 2010.
2. Ganguli, Prabuddha. *Intellectual Property Rights: Unleashing the Knowledge Economy*. Tata McGraw Hill, 2001.
3. Dhar, Biswajit, and Chaturvedi, Sachin. *Regulating Access to Biological Resources and Benefit Sharing: Challenges for India*. RIS, New Delhi, 2006.
4. Dutfield, Graham. *Protecting Traditional Knowledge: Pathways to the Future*. International Centre for Trade and Sustainable Development (ICTSD), Geneva, 2006.
5. Basu, Indira. *Traditional Knowledge and Intellectual Property Rights*. Deep & Deep Publications, New Delhi, 2009.

Articles and Journals:

1. Suneetha, M.S., and Pisupati, Balakrishna. "Benefit Sharing in ABS: Options and Implications for India." *Asian Biotechnology and Development Review*, Vol. 8, No. 2, 2006.
2. Sharma, R.K. "Protection of Traditional Knowledge in India: Legal and Policy Perspectives." *Journal of Intellectual Property Rights*, Vol. 21, 2016, pp. 245–253.
3. Gupta, V. "Traditional Knowledge and Biopiracy: Legal Challenges and the Indian Approach." *Indian Journal of Law and Technology*, Vol. 11, 2015, pp. 123–137.
4. Kumar, S. "Traditional Knowledge and the Role of TKDL in India." *Journal of World Intellectual Property*, Vol. 22, No. 3, 2019.
5. Ramesh, G. "Protection of Indigenous Knowledge under IPR Regime." *Indian Law Review*, Vol. 12, 2018, pp. 88–104.

Reports and Government Documents:

1. Government of India, *The Biological Diversity Act, 2002*. Ministry of Environment, Forest and Climate Change.
2. Ministry of Science and Technology, *Traditional Knowledge Digital Library (TKDL) Annual Report*, CSIR, Government of India, 2020.
3. WIPO, *Intellectual Property and Genetic Resources, Traditional Knowledge and Traditional Cultural Expressions*, World Intellectual Property Organization, Geneva, 2017.

4. National Biodiversity Authority, *Guidelines for Access and Benefit Sharing in India*, Government of India, 2014.
5. Planning Commission of India, *Report of the Task Force on Traditional Knowledge, Innovation and Practices*, 2000.

Cases:

1. *Neem Case*: W.R. Grace & Co. v. Indian Council of Agricultural Research (European Patent Office, 2000).
2. *Turmeric Case*: Re Turmeric Patent (USPTO, 1997).
3. *Basmati Rice Case*: RiceTec Inc. Patent Controversy (USPTO, 2001).
4. *Arogya Pacha Case*: Kani Tribe and Tropical Botanical Garden & Research Institute Benefit Sharing Agreement (India, 2000).

Websites and Online Resources:

1. *Traditional Knowledge Digital Library (TKDL)*, Council of Scientific and Industrial Research (CSIR), Government of India, available at: <https://www.tkdil.res.in> (last visited Nov. 8, 2025).
 2. *World Intellectual Property Organization (WIPO) – Traditional Knowledge Portal*, available at: <https://www.wipo.int/tk/en/> (last visited Nov. 8, 2025).
 3. *National Biodiversity Authority, India*, Ministry of Environment, Forest and Climate Change, available at: <https://nbaindia.org> (last visited Nov. 8, 2025).
 4. *Ministry of Environment, Forest and Climate Change, Government of India*, available at: <https://moef.gov.in> (last visited Nov. 8, 2025).
 5. *Convention on Biological Diversity (CBD)*, United Nations, available at: <https://www.cbd.int> (last visited Nov. 8, 2025).
 6. *United Nations Environment Programme (UNEP)*, available at: <https://www.unep.org> (last visited Nov. 8, 2025).
-
1. Anil K. Gupta, *Intellectual Property Rights and Traditional Knowledge: The Indian Experience* (Sage Publications, New Delhi, 2010) 45.
 2. Graham Dutfield, *Protecting Traditional Knowledge: Pathways to the Future* (International Centre for Trade and Sustainable Development, Geneva, 2006) 12.
 3. Ministry of Science and Technology, *Traditional Knowledge Digital Library (TKDL) Annual Report* (Council of Scientific and Industrial Research, Government of India, New Delhi, 2020) 8.
 4. Council of Scientific and Industrial Research (CSIR), *Traditional Knowledge Digital Library (TKDL): Protecting India's Traditional Knowledge* (Government of India, New Delhi, 2019) 5.
 5. Government of India, *The Biological Diversity Act, 2002* (Ministry of Environment, Forest and Climate Change, New Delhi, 2002) 3.
 6. Biswajit Dhar and Sachin Chaturvedi, *Regulating Access to Biological Resources and Benefit Sharing: Challenges for India* (Research and Information System for Developing Countries, New Delhi, 2006) 14.
 7. P. Pushpangadan and C.K. Atal, *Traditional Medicine: Modern Approach for Affordable Global Health* (National Botanical Research Institute, Lucknow, 2005) 22.
 8. Vandana Shiva, *Staying Alive: Women, Ecology and Development* (Zed Books, London, 1989) 67.
 9. Kapila Vatsyayan, *Traditional Indian Arts: The Cultural Context* (Publication Division, Ministry of Information and Broadcasting, Government of India, New Delhi, 2003) 41.
 10. Madhav Gadgil and Fikret Berkes, *Traditional Resource Management Systems* (Centre for Ecological Sciences, Indian Institute of Science, Bangalore, 1991) 29.
 11. V.K. Gupta, *Traditional Knowledge Digital Library: A Step Towards Protecting India's Traditional Knowledge* (Council of Scientific and Industrial Research, New Delhi, 2002) 10.
 12. Anil K. Gupta, *Rewarding Local Innovation: Traditional Knowledge and Sustainable Development* (Indian Institute of Management, Ahmedabad, 1999) 18. collective, old, and come from many people, but modern law looks for one inventor and new creation. This mismatch makes it difficult to give legal protection.¹³
 13. Nirmal Sengupta, *Traditional Knowledge and Intellectual Property Rights: Conflict and Prospects* (Indian National Science Academy, New Delhi, 2007) 24.
 14. R.S. Chauhan, *Protection of Traditional Knowledge and Biopiracy in India* (Journal of Intellectual Property Rights, Vol. 15, 2010) 213.
 15. Shammad Basheer, *Protecting Traditional Knowledge in India: Towards a Sui Generis Framework* (Journal of National Law University, Delhi, Vol. 3, No. 1, 2011) 56.
 16. Ritu Priya and S.K. Sharma, *Traditional Healers and Knowledge Holders: Need for Awareness and Legal Protection* (Indian Journal of Traditional Knowledge, Vol. 15, No. 4, 2016) 589.
 17. UNESCO, *Local and Indigenous Knowledge Systems: Protecting Knowledge for Sustainable Development* (UNESCO, Paris, 2018) 12.

18. WIPO, Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore: Report on Traditional Knowledge and Intellectual Property (World Intellectual Property Organization, Geneva, 2020) 8.
19. The Biological Diversity Act, 2002 (Act No. 18 of 2003), Gazette of India, Ministry of Environment and Forests, Government of India.
20. Traditional Knowledge Digital Library (TKDL), Council of Scientific and Industrial Research (CSIR), Government of India, New Delhi, available at: <https://www.tkdil.res.in> (last visited Nov. 8, 2025). make sure that people who hold it get fair share when it is used. These agreements help India to make its national law strong and connected with world system.²¹ Other law like Patents Act, 1970, Plant Varieties and Farmers' Rights Act, 2001, and Forest Rights Act, 2006 also give some protection to local people who use and keep traditional knowledge. The government has also made policy and program to support documentation and awareness among people.²²
21. Convention on Biological Diversity (CBD), 1992, and Nagoya Protocol on Access and Benefit Sharing, 2010, United Nations, available at: <https://www.cbd.int> (last visited Nov. 8, 2025).
22. The Patents Act, 1970; The Protection of Plant Varieties and Farmers' Rights Act, 2001; and The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Government of India, available at: <https://legislative.gov.in> (last visited Nov. 8, 2025).
23. Convention on Biological Diversity, 1992, United Nations, available at: <https://www.cbd.int/convention/> (last visited Nov. 8, 2025).
24. Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization, 2010, United Nations, available at: <https://www.cbd.int/abs/> (last visited Nov. 8, 2025).
25. WIPO, Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore: Annual Report 2020 (World Intellectual Property Organization, Geneva, 2020) 5.
26. W.R. Grace & Co. v. Indian Council of Agricultural Research, European Patent Office, Case No. 436 257 (2000).
27. Re Turmeric Patent (USPTO, 1997).
28. RiceTec Inc. Patent Controversy, United States Patent and Trademark Office (USPTO), 2001.
29. Vandana Shiva, Biopiracy: The Plunder of Nature and Knowledge (South End Press, Boston, 1997) 34.
30. Ritu Priya and S.K. Sharma, Traditional Healers and Knowledge Holders: Need for Awareness and Legal Protection (Indian Journal of Traditional Knowledge, Vol. 15, No. 4, 2016) 592.
31. Madhav Gadgil and Fikret Berkes, Traditional Resource Management Systems (Centre for Ecological Sciences, Indian Institute of Science, Bangalore, 1991) 35.
32. Shamnad Basheer, Protecting Traditional Knowledge in India: Towards a Sui Generis Framework (Journal of National Law University, Delhi, Vol. 3, No. 1, 2011) 58.
33. Ministry of Environment, Forest and Climate Change, Guidelines for Access and Benefit Sharing in India (National Biodiversity Authority, Government of India, 2014) 7.
34. The Biological Diversity Act, 2002 (Act No. 18 of 2003), Gazette of India, Ministry of Environment, Forest and Climate Change, Government of India.
35. Suneetha, M.S., and Balakrishna Pisupati, "Benefit Sharing in ABS: Options and Implications for India," Asian Biotechnology and Development Review, Vol. 8, No. 2, 2006, 45.
36. Shamnad Basheer, Protecting Traditional Knowledge in India: Towards a Sui Generis Framework (Journal of National Law University, Delhi, Vol. 3, No. 1, 2011) 61.
37. Ministry of Environment, Forest and Climate Change, Guidelines for Access and Benefit Sharing in India (National Biodiversity Authority, Government of India, 2014) 9.
38. WIPO, Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore: Annual Report 2020 (World Intellectual Property Organization, Geneva, 2020) 12.